COMP3016 / COURSEWORK 1 PROPOSAL

Miracle Author

Danial Afzan | Single-Player 2D Platformer Game

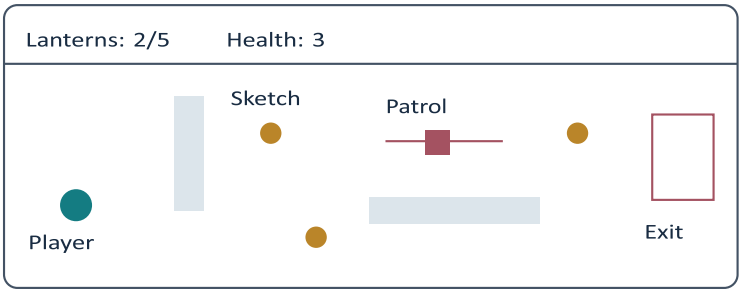
Concept & core loop. An aspiring artist wanting to paint the vibrant city of New York before her deadline. Explore the city, meet the requirements and avoid unwanted attention from patrolling officers.

MDA: MECHANICS / DYNAMICS / AESTHETICS

Mechanics (rules): WASD/arrow-key movement; solid walls with wall jumping mechanics ; find and trace back pin points of the city using reference and hints you've gathered; patrolling enemies following a fixed route; three health points. Contact costs one health point, followed by one second of invulnerability. Next area unlocks after completing the objective in the current level. Zero health ends the run.

Dynamics (player behaviour): Patrols and limited health encourage observation and careful timing. Wall jumping mechanics makes players find safer detours and shorter, riskier routes. Collecting the last sketch changes the objective from exploration to escape; restarting supports learning a better route.

Aesthetics (intended experience): Challenge through timed navigation, tension near enemies, and satisfaction from completing the deadline. Colorful sketches of the city colours against the city of the night create a vibrant view. Clear pickup feedback and a visible exit state make progress easy to understand.



GAMEPLAY SKETCH & OOP STRUCTURE

Class sketch (owns / manages)

Game

Player: move(), takeDamage()

Level: load(), walls, gate

Enemy[]: patrol(), update()

Collectible[]: collect()

Game controls input, updates, rendering and Menu / Playing / Win / Lose states.

TECHNICAL PLAN & SCOPE

C++ / SDL3: encapsulated classes, standard containers, delta-time movement and rectangle collision checks. Load walls, spawn positions and sketch locations from a plain-text level file at runtime; validate missing or malformed data. Use SDL3 for rendering/input and optional WAV audio; no additional external libraries planned. MVP uses simple original shapes. **Stretch only:** sprite animation, sound and a second level.

GIT MILESTONES & VALIDATION

22-25 Oct: repository/setup, window, input, Player and file-loaded arena; commit evidence: "Load level and implement player movement".
26 Oct-1 Nov: collisions, pickups, patrols, timer, health and game states; deliver a complete playable loop, with separate feature commits.
2-5 Nov: peer playtest, record issues, refine difficulty, test restart and invalid files, check build on an SMB102 PC; finish README, AI log and demo video.
6 Nov: final packaging and submission before 15:00 (brief deadline). Commit each meaningful change; record screenshots, test results and justified scope changes in GitHub.

Success checks: blocked walls; each pickup counts once; locked gate until 5/5; correct timeout/health loss; restart resets all state. Ask peers to complete a run and use their feedback to adjust patrol speed or the timer.
AI declaration: ChatGPT assisted this proposal and concept. Any later AI support for debugging, refactoring or assets will be logged with prompts, checks and modifications; all submitted code will be understood and tested.